DISCRETE STRUCTURES

CMSC250

Chapter 6: Predicates and Quantifiers

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Turning statements about objects into precise logic



Notation. Negation is written $\sim$ (also commonly written $\neg$). The symbols $\forall$ and $\exists$ mean "for every" and "there exists." Domains must be specified.

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1 Why propositional logic is not enough

Propositional logic treats a whole sentence as one unit. If p means "Every student passed," then p can be true or false, but the symbol p does not reveal *which* students are involved. To describe individual objects and make claims about all or some of them, we use **predicate logic**.

For example, let $P(x)$ mean "x passed the quiz." Then $P(\text{Ava})$ is a statement about one student, while $\forall x\, P(x)$ is a statement about every student in a specified domain. The difference is important: a predicate contains a variable whose value has not yet been fixed.

Key idea. A proposition already has a truth value. A predicate such as $P(x)$ becomes a proposition when its variable is assigned a value or bound by a quantifier (assuming the interpretation is fixed).

2 Predicates, variables, and domains

A **predicate** is a property or relationship expressed using variables. Examples include

$$E(x):\ \text{“}x \text{ is even”}, \qquad L(x,y):\ \text{“}x < y\text{”}.$$

The **domain of discourse** specifies which values the variables may take. A formula can change truth value when its domain changes.

Worked example: one formula, different domains

Let $P(x)$ mean $x^2 \geq x$. Over $\mathbb{N} = \{0, 1, 2, \ldots\}$, $P(x)$ is true for every x. Over $\mathbb{R}$, it is not: $x = \frac{1}{2}$ gives $\frac{1}{4} \not\geq \frac{1}{2}$. Therefore, writing "$\forall x\, P(x)$" without stating the domain can be ambiguous.

A **unary** predicate has one argument, such as $E(x)$. A **binary** predicate has two, such as $L(x, y)$. The argument order matters: $L(2, 5)$ is true while $L(5, 2)$ is false.

3 The universal quantifier $\forall$

The formula $\forall x \in D,\ P(x)$ means that **every** element of D satisfies P. To establish it for a finite domain, check every element; to disprove it, find a single counterexample.

$$\forall x \in \{1, 2, 3\},\ x < 4$$

is true because $1 < 4$, $2 < 4$, and $3 < 4$ are all true. In contrast, $\forall x \in \mathbb{Z},\ x > 0$ is false: $x = 0$ is a counterexample.

Key idea. To disprove a universal statement, one counterexample is enough. To prove a universal statement, one example is never enough.

4 The existential quantifier $\exists$

The formula $\exists x \in D,\ P(x)$ means that **at least one** element of D satisfies P. To prove it, exhibit a valid witness. To disprove it, show that no element of the domain works.

Worked example: a witness versus a counterexample

Over the integers, $\exists x\,(x^2 = 9)$ is true: $x = 3$ is a witness (and $x = -3$ is another). But $\exists x \in \mathbb{Z}\,(x^2 = 2)$ is false: no integer has square 2. The word "exists" does not mean "exactly one."

5 Reading quantifiers over finite domains

For $D = \{a, b, c\}$, universal and existential statements can be expanded into propositional logic:

$$\forall x \in D\, P(x) \;\equiv\; P(a) \wedge P(b) \wedge P(c),$$

$$\exists x \in D\, P(x) \;\equiv\; P(a) \vee P(b) \vee P(c).$$

This explains why universal claims behave like a long AND and existential claims behave like a long OR.

Worked example: evaluating a finite interpretation

Let $D = \{1, 2, 3, 4\}$ and $E(x)$ mean "x is even." Then $\forall x \in D\, E(x)$ is false (for instance, 1 is not even), but $\exists x \in D\, E(x)$ is true (witness 2). Neither result depends on how many additional witnesses exist.

6 Negating quantified statements

The negation of "everyone has property P" is "someone does not have property P." The negation of "someone has property P" is "everyone lacks property P." Formally,

$$\boxed{\sim (\forall x \in D\, P(x)) \;\equiv\; \exists x \in D \;\sim P(x)}$$

$$\boxed{\sim (\exists x \in D\, P(x)) \;\equiv\; \forall x \in D \;\sim P(x)}.$$

Important: changing the quantifier is not enough; the predicate must also be negated.

Worked example: negating an English sentence

Original: "Every student submitted the assignment." Define $S(x)$ as "student x submitted the assignment," with students as the domain. The formula is $\forall x\, S(x)$. Its negation is $\exists x \sim S(x)$: "At least one student did not submit the assignment." It is *not* "No student submitted the assignment."

Worked example: negating an existential statement

Original: "Some integer is greater than every integer." Its negation is "Every integer has an integer at least as large as it" only if phrased with the proper relational structure; explicitly, if the original is $\exists x \in \mathbb{Z}\, \forall y \in \mathbb{Z}\, (x > y)$, then the exact negation is

$$\forall x \in \mathbb{Z}\, \exists y \in \mathbb{Z}\, (x \leq y).$$

We changed both quantifiers and negated $>$ to $\leq$.

7 Nested quantifiers and their order

When a predicate uses more than one variable, quantifiers may be nested. The order is sometimes decisive:

$$\forall x \in D\, \exists y \in D\, R(x, y) \qquad \text{versus} \qquad \exists y \in D\, \forall x \in D\, R(x, y).$$

The first permits a different y for each x; the second requires one *fixed* y that works for all x.

Worked example: two orders, different truth values

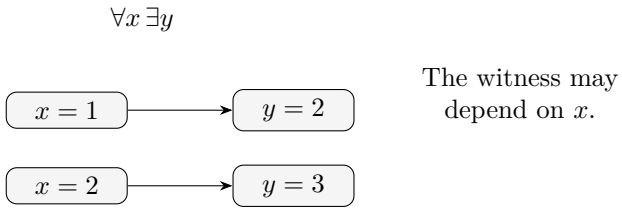
Let $D = \mathbb{Z}$ and $R(x, y)$ mean $y > x$. Then

$$\forall x \in \mathbb{Z}\, \exists y \in \mathbb{Z}\, (y > x)$$

is true: given any integer x, choose $y = x + 1$. However,

$$\exists y \in \mathbb{Z}\, \forall x \in \mathbb{Z}\, (y > x)$$

is false: no single integer is greater than every integer. Given a proposed y, choose $x = y + 1$.



7.1 When can quantifiers be swapped?

Two universal quantifiers commute: $\forall x\, \forall y\, P(x, y)$ is equivalent to $\forall y\, \forall x\, P(x, y)$. Two existential quantifiers also commute. Mixed quantifiers generally *cannot* be swapped without changing the claim.

8 Free and bound variables

A variable is **bound** when it is controlled by a quantifier within that quantifier's scope. A variable is **free** if no quantifier binds it.

In $\forall x\, (P(x, y) \to Q(x))$, the occurrences of x are bound, but y is free. The expression does not yet determine a truth value without specifying y (or binding it with another quantifier).

Worked example: finding free variables

Consider $\exists y\, (R(x, y) \wedge S(y, z))$. Here y is bound, while x and z are free. If we prepend $\forall x\, \forall z$, all variables become bound, producing a closed statement.

9 Translating English into predicate logic

Choose a domain first, define predicates clearly, and then translate the quantifier words. For students as the domain, let $C(x)$ mean "x completed the project" and $T(x)$ mean "x passed the test."

English	Logical form
Every student completed the project.	$\forall x\, C(x)$
At least one student completed it.	$\exists x\, C(x)$
No student completed it.	$\forall x\ \sim C(x)$
Some student completed it but failed the test.	$\exists x\, (C(x) \wedge \sim T(x))$
Every student who completed it passed the test.	$\forall x\, (C(x) \rightarrow T(x))$
Some student who completed it passed the test.	$\exists x\, (C(x) \wedge T(x))$

9.1 Why universal restrictions use implication

Suppose the domain is all people, $S(x)$ means "x is a student," and $P(x)$ means "x passed." "Every student passed" becomes

$$\forall x\, (S(x) \rightarrow P(x)).$$

People who are not students do not violate this claim. Writing $\forall x\, (S(x) \wedge P(x))$ would incorrectly assert that *everyone is a student.*

9.2 Why existential restrictions use conjunction

"Some student passed" becomes $\exists x\, (S(x) \wedge P(x))$. We need one person who satisfies *both* properties. The formula $\exists x\, (S(x) \rightarrow P(x))$ is too weak: any nonstudent makes the implication true.

Key idea. For "every A is B," use $\forall x\, (A(x) \rightarrow B(x))$. For "some A is B," use $\exists x\, (A(x) \wedge B(x))$, when the domain includes objects outside A.

10 At least one, at most one, and exactly one

Let $P(x)$ mean "x has the desired property."

$$\begin{aligned} &\text{At least one:} && \exists x\, P(x). \\ &\text{At most one:} && \forall x\, \forall y\, ((P(x) \wedge P(y)) \rightarrow x = y). \\ &\text{Exactly one:} && \exists x\, (P(x) \wedge \forall y\, (P(y) \rightarrow y = x)). \end{aligned}$$

The at-most-one statement permits *zero* objects with the property. Exactly one combines existence with uniqueness.

Worked example: exactly one student received full credit

With students as the domain and $F(x)$ meaning "x received full credit," write

$$\exists x\, (F(x) \wedge \forall y\, (F(y) \rightarrow y = x)).$$

The first part guarantees a student received full credit. The second says any student who received full credit must be that same student.

11 Interpreting statements and finding counterexamples

An **interpretation** specifies a domain and assigns meanings to predicates. To show that a formula is *not universally valid*, it suffices to give one interpretation where it is false.

Worked example: a counterinterpretation

Compare $\forall x\,(P(x) \lor Q(x))$ with $(\forall x\,P(x)) \lor (\forall x\,Q(x))$. Take $D = \{1, 2\}$, let $P(1)$ be true and $P(2)$ false, and let $Q(1)$ be false and $Q(2)$ true. Every element satisfies P or Q, so the first formula is true. But not every element satisfies P, and not every element satisfies Q, so the second formula is false. Thus the two formulas are not equivalent.

12 Empty domains and vacuous truth

When the domain is empty, a universal statement $\forall x \in \varnothing\, P(x)$ is true: there is no counterexample. An existential statement $\exists x \in \varnothing\, P(x)$ is false: there is no witness. This convention is consistent with universal quantification behaving like conjunction and existential quantification behaving like disjunction.

Some courses implicitly assume domains are nonempty. When a problem explicitly permits an empty domain, the distinction matters and should be stated.

13 Common mistakes and grading pitfalls

1. **Changing $\forall$ to $\exists$ accidentally.** "Every student" and "some student" are logically different; this is not a harmless notation slip.
2. **Negating only the quantifier.** Both the quantifier and the inside predicate must change when pushing negation inward.
3. **Swapping mixed quantifiers.** A different witness for every object is not the same as one universal witness.
4. **Forgetting the domain.** The same formula may be true over $\mathbb{N}$ but false over $\mathbb{R}$.
5. **Using AND for a universal restriction.** "All students pass" does not mean "everyone is a student and passes."
6. **Using implication for an existential restriction.** A nonstudent can make $S(x) \to P(x)$ true without supplying a student who passed.
7. **Confusing "at most one" with "exactly one."** At most one permits no witnesses.
8. **Providing only examples for a universal proof.** A finite sample cannot establish a claim over an infinite domain.

14 Practice problems

Use the domains stated in each question. Work out the answers before consulting the solutions.

1. Let $D = \{0, 1, 2\}$ and $P(x)$ mean $x^2 = x$. Determine the truth values of $\forall x \in D\, P(x)$ and $\exists x \in D\, P(x)$.

2. Over $\mathbb{Z}$, determine whether $\forall x\, \exists y\, (x + y = 0)$ is true. Justify your answer.

3. Over $\mathbb{Z}$, determine whether $\exists y\, \forall x\, (x + y = 0)$ is true. Justify your answer.

4. Negate $\forall x \in \mathbb{R}\, (x^2 \geq 0)$, moving $\sim$ inward.

5. Negate $\exists x \in \mathbb{Z}\, \forall y \in \mathbb{Z}\, (x \leq y)$.

6. With all people as the domain, let $S(x)$ mean "x is a student" and $R(x)$ mean "x reads." Translate "Every student reads."

7. Using the predicates from Problem 6, translate "At least one student does not read."

8. Let $L(x, y)$ mean "x likes y," with people as the domain. Translate "Everyone likes someone."

9. Using L, translate "Someone is liked by everyone." Explain why the order matters.

10. Identify all free variables in $\forall x\, (P(x, y) \vee \exists z\, R(z, w))$.

11. Write a formula expressing "At most one student passed," with students as the domain and $P(x)$ meaning "x passed."

12. Write a formula expressing "Exactly one student passed," using the same domain and predicate.

13. Let $D = \{a, b\}$, $P(a) = 1$, $P(b) = 0$. Expand $\forall x \in D\, P(x)$ into propositional logic and evaluate it.

14. Construct an interpretation in which $\exists x\, (P(x) \vee Q(x))$ is true but $(\exists x\, P(x)) \wedge (\exists x\, Q(x))$ is false.

15. Explain the truth values of $\forall x \in \varnothing\, P(x)$ and $\exists x \in \varnothing\, P(x)$.

16. Over $\mathbb{Z}$, decide whether $\forall x\, \exists y\, (y = x + 1)$ and $\exists y\, \forall x\, (y = x + 1)$ are true or false.

15 Detailed solutions

1. $P(0)$ and $P(1)$ are true, but $P(2)$ is false since $4 \neq 2$. Therefore the universal statement is false and the existential statement is true.

2. True. Given an arbitrary integer x, choose $y = -x$, which is an integer. Then $x + y = 0$.

3. False. A single y would need to equal 0 when $x = 0$ and -1 when $x = 1$, which is impossible.

4. $\exists x \in \mathbb{R}\, (x^2 < 0)$. The negated statement is false, since squares of real numbers are nonnegative.

5. $\forall x \in \mathbb{Z}\, \exists y \in \mathbb{Z}\, (x > y)$. The negation changes both quantifiers and reverses $\leq$ to $>$.

6. $\forall x\, (S(x) \to R(x))$. Nonstudents do not violate the claim.

7. $\exists x\, (S(x) \wedge \sim R(x))$. One witness must be both a student and a nonreader.

8. $\forall x\,\exists y\,L(x,y)$. The liked person may depend on who x is.

9. $\exists y\,\forall x\,L(x,y)$. One fixed person y must be liked by everyone, which is stronger than allowing a different person for each x.

10. y and w are free. The quantifiers bind x and z.

11. $\forall x\,\forall y\,((P(x)\wedge P(y))\to x=y)$. It is also true if nobody passed.

12. $\exists x\,(P(x)\wedge\forall y\,(P(y)\to y=x))$. Existence and uniqueness are both required.

13. $P(a)\wedge P(b)=1\wedge 0=0$. The universal statement is false.

14. Take $D=\{a\}$ with $P(a)$ true and $Q(a)$ false. Then $\exists x\,(P(x)\vee Q(x))$ is true, but $\exists x\,Q(x)$ is false, so the conjunction is false.

15. The universal formula is true vacuously because there is no counterexample; the existential formula is false because there is no witness.

16. The first is true: for any x, choose $y=x+1$. The second is false: one fixed y cannot equal both 1 (when $x=0$) and 2 (when $x=1$).